Grade 12 Advanced Functions
Logarithmic Functions Key

First Name: _____**KEY**_____

Last Name: _____

Directions:

- Please answer to 2 decimal points
- The test is designed to be completed in 75 minutes

For grading use only

Page:	2	3	4	5	Total
Points:	6	18	9	9	42
Score:					

Multiple Choice (6 marks)

1. (1 point) Evaluate $\log_2(32)$.
   - **A.** 5
   - B. 4
   - C. 3
   - D. 2

2. (1 point) Rewrite $5^3 = 125$ in logarithmic form.
   - **A.** $\log_5(125) = 3$
   - B. $\log_{125}(5) = 3$
   - C. $\log_3(5) = 125$
   - D. $\log(5) = 3$

3. (1 point) Simplify using laws of logs (assume $x > 0$): $\log(2x) + \log(5)$.
   - A. $\log\left(\frac{2x}{5}\right)$
   - **B.** $\log(10x)$
   - C. $\log(7x)$
   - D. $\log(2) + \log(x) + 5$

4. (1 point) Solve for x (domain enforced): $\log_3(x+1) = 2$.
   - **A.** $x = 8$
   - B. $x = 9$
   - C. $x = 7$
   - D. $x = 3$

5. (1 point) Solve for x: $4^x = 64$.
   - A. 2
   - **B.** 3
   - C. 4
   - D. 6

6. (1 point) For $y = \log_2(x-3) + 1$, the vertical asymptote is:
   - **A.** $x = 3$
   - B. $x = 0$
   - C. $y = 1$
   - D. $x = -3$

Long Answer (36 marks)

7. Laws and Evaluation of Logarithms (9 marks) Let all logs be base 10 unless otherwise stated and assume variables are in their natural domains.
   (a) (4 points) Simplify fully: $\log(2x) + \log(5) - \frac{1}{2}\log(x^2)$. State the domain.
   (b) (3 points) Expand $\ln\left(\dfrac{9x^2\sqrt{y}}{3(y-1)^3}\right)$ into a sum/difference of logs (no powers inside logs).
   (c) (2 points) Evaluate $\log_2\left(\dfrac{1}{8}\right)$ and rewrite $7^2 = 49$ as a logarithmic statement.

Solution: (a) Use product/quotient/power rules: $\log A + \log B = \log(AB)$**,** $\log A - \log B = \log\left(\frac{A}{B}\right)$**,** $k\log A = \log(A^k)$**.**

$$\log(2x) + \log(5) = \log(10x), \quad -\tfrac{1}{2}\log(x^2) = \log\big((x^2)^{-1/2}\big) = \log\left(\frac{1}{|x|}\right).$$

Thus

$$\log(10x) - \log(x) = \log\left(\frac{10x}{x}\right) = \log(10) = 1,$$

but be careful: using $\log\left(\frac{1}{|x|}\right)$ **explicitly,**

$$\log(10x) + \log\left(\frac{1}{|x|}\right) = \log\left(\frac{10x}{|x|}\right) = \log(10) = 1.$$

So the simplified value is 1**. Domain: for original expression, need** $x > 0$ **to define** $\log(2x)$ **and** $\log(x^2)$ **consistently with the manipulation to** $\log x$ **(and also ensure** $|x| = x$**). Hence** $x > 0$**.**

(b) Write each factor with separate logs and apply power rule:

$$\ln\left(\frac{9x^2\sqrt{y}}{3(y-1)^3}\right) = \ln(9) + \ln(x^2) + \ln(y^{1/2}) - \ln(3) - \ln\big((y-1)^3\big).$$

Now simplify constants and exponents:

$$\ln(9) - \ln(3) = \ln\left(\frac{9}{3}\right) = \ln(3), \quad \ln(x^2) = 2\ln x, \quad \ln(y^{1/2}) = \tfrac{1}{2}\ln y, \quad \ln\big((y-1)^3\big) = 3\ln|y-1|.$$

Therefore

$$\boxed{\ln(3) + 2\ln x + \tfrac{1}{2}\ln y - 3\ln|y-1|}.$$

Domain: $x > 0$**,** $y > 0$**, and** $y \neq 1$**.**

(c) $\log_2\left(\dfrac{1}{8}\right) = \log_2(2^{-3}) = -3$**. Also** $7^2 = 49$ **becomes** $\log_7(49) = 2$**.**

8. Solving Logarithmic and Exponential Equations (9 marks)
   (a) (4 points) Solve $\log_3(x+1) = 2$. Show domain checks.
   (b) (3 points) Solve $2\log_5(x) - \log_5(4) = 1$ for $x > 0$.
   (c) (2 points) Solve $4^x = 64$ by expressing each side with the same base.

Solution: (a) Start from the definition $\log_a(b) = c \iff a^c = b$ **(with** $a > 0, a \neq 1, b > 0$**).**

$$\log_3(x+1) = 2 \iff x+1 = 3^2 = 9 \iff x = 8.$$

Domain check: $x + 1 > 0 \Rightarrow x > -1$**. The solution** $x = 8$ **satisfies this. Answer:** $x = 8$**.**

(b) Combine logs into a single logarithm:

$$2\log_5(x) - \log_5(4) = \log_5(x^2) - \log_5(4) = \log_5\left(\frac{x^2}{4}\right) = 1.$$

Rewrite in exponential form:

$$\log_5\left(\frac{x^2}{4}\right) = 1 \iff \frac{x^2}{4} = 5^1 = 5 \iff x^2 = 20 \iff x = \pm 2\sqrt{5}.$$

Domain: $\log_5(x)$ **requires** $x > 0$**. Therefore** $x = 2\sqrt{5}$**.**

(c) Express with base 4**:** $64 = 4^3$**, hence**

$$4^x = 4^3 \Rightarrow x = 3.$$

9. Transformations and Graph Features (9 marks) For $f(x) = \log_2(x-3) + 1$:
   (a) (3 points) State the domain, range, and vertical asymptote.
   (b) (3 points) Find the x- and y-intercepts (if any), or explain if none exist.
   (c) (3 points) Describe the transformation(s) from $y = \log_2(x)$ and indicate two additional points on the graph besides any intercepts (compute exact values).

Solution: (a) Inside the log must be positive: $x - 3 > 0 \Rightarrow x > 3$**. So**

$$\textbf{Domain } (3, \infty), \quad \textbf{vertical asymptote } x = 3.$$

For any logarithm, the range is all real numbers. Hence Range $(-\infty, \infty)$**.**

(b) y**-intercept: requires** $x = 0$**, but** $0 \notin (3, \infty)$**, so no** y**-intercept. For the** x**-intercept, set** $y = 0$**:**

$$0 = \log_2(x-3) + 1 \iff \log_2(x-3) = -1 \iff x - 3 = 2^{-1} = \frac{1}{2} \Rightarrow x = \frac{7}{2}.$$

Thus intercept at $\left(\frac{7}{2}, 0\right)$**.**

(c) From $y = \log_2(x)$**: shift right by** 3 **units (replace** x **with** $x - 3$**), then shift up by** 1 **unit (add** $+1$**). Choose easy inputs where** $x - 3$ **is a power of** 2**:**

$$x - 3 = 1 \Rightarrow x = 4 \Rightarrow f(4) = \log_2(1) + 1 = 0 + 1 = 1 \quad \Rightarrow (4, 1).$$

$$x - 3 = 2 \Rightarrow x = 5 \Rightarrow f(5) = \log_2(2) + 1 = 1 + 1 = 2 \quad \Rightarrow (5, 2).$$

These points, together with the asymptote $x = 3$ **and the** x**-intercept** $\left(\frac{7}{2}, 0\right)$**, determine the sketch.**

10. Applications of Logarithms (Real-World) (9 marks)

(a) (4 points) Continuous compounding: How long will it take an investment of \$1000 to double at 5% per year, compounded continuously? Use $A(t) = Pe^{rt}$ and solve for t.

(b) (5 points) A radioactive substance decays according to $N(t) = N_0 e^{-kt}$. If a sample decreases to 60% of its initial amount in 5 years, find k and the half-life (time to reach 50%). Give exact forms then approximations.

Solution: (a) Set $A = 2P$ with $r = 0.05$:

$$2P = Pe^{0.05t} \;\Rightarrow\; 2 = e^{0.05t}.$$

Take natural logs (since $\ln$ is inverse of $e^{\cdot}$):

$$\ln 2 = 0.05\,t \;\Rightarrow\; t = \frac{\ln 2}{0.05} \approx \frac{0.693147}{0.05} \approx 13.8629 \text{ years}.$$

So doubling time is about 13.86 years.

(b) Use $N(t) = N_0 e^{-kt}$. Given $N(5) = 0.6N_0$:

$$0.6N_0 = N_0 e^{-5k} \;\Rightarrow\; 0.6 = e^{-5k}.$$

Take $\ln$:

$$\ln(0.6) = -5k \;\Rightarrow\; k = -\frac{1}{5}\ln(0.6) = \frac{1}{5}\ln\left(\frac{1}{0.6}\right) = \frac{1}{5}\ln\left(\frac{5}{3}\right).$$

Exact: $k = \frac{1}{5}\ln\left(\frac{5}{3}\right) \approx \frac{1}{5}\cdot 0.510826 \approx 0.102165$ **(per year).**

Half-life $T_{1/2}$ satisfies $N(T_{1/2}) = \frac{1}{2}N_0$:

$$\frac{1}{2} = e^{-kT_{1/2}} \;\Rightarrow\; \ln\left(\frac{1}{2}\right) = -kT_{1/2} \;\Rightarrow\; T_{1/2} = \frac{\ln 2}{k}.$$

Using exact k:

$$T_{1/2} = \frac{\ln 2}{\frac{1}{5}\ln\left(\frac{5}{3}\right)} = \frac{5\ln 2}{\ln\left(\frac{5}{3}\right)} \approx \frac{5\cdot 0.693147}{0.510826} \approx \frac{3.46574}{0.510826} \approx 6.787 \text{ years}.$$

Thus $k \approx 0.1022$ yr^{-1} and half-life ≈ 6.79 years.